International Journal of Hydrogen Energy 59 (2024) 51–62

Contents lists available at ScienceDirect

International Journal of Hydrogen Energy

journal homepage: www.elsevier.com/locate/he

Parametric analysis of wastewater electrolysis for green hydrogen production: A combined RSM, genetic algorithm, and particle swarm optimization approach

Aqueel Ahmad [a,*], Ashok Kumar Yadav [b]

[a] Mechanical Engineering Department, Netaji Subhas University of Technology, New Delhi, India
[b] Department of Mechanical Engineering, Raj Kumar Goel Institute of Technology, Ghaziabad, India

ARTICLE INFO

Handling editor: J Ortiz

Keywords:
Electrolysis
HHO
Prediction
Optimization
RSM
GA
PSO

ABSTRACT

The production of green hydrogen from wastewater presents a significant opportunity to address environmental challenges associated with wastewater treatment and fulfill the increasing demand for renewable and sustainable energy sources. This study aimed to determine the optimal operating parameters for maximizing hydrogen production through wastewater electrolysis. To design the experimental runs, a Box-Behnken design (BBD) with an L_{17} array was implemented, while Response Surface Methodology (RSM) in conjunction with Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) was employed to develop predictive models and optimize the operating parameters. The operating parameters considered were the catalyst amount (g), electrode voltage (V), and electrolysis time (min), and the response variable was the oxyhydrogen (HHO) gas generation rate (L/min). The comparative analysis shows that the optimized parameters obtained through RSM surpassed those obtained through GA and PSO, with a catalyst amount of 99.42 g, an electrode voltage of 22.9 V, and an electrolysis time of 23.17 min. Consequently, the HHO generation rate reached a maximum value of 12.42 L/min. Furthermore, the experimental validation indicated a close agreement between the model's predicted results and the actual experimental outcomes. This study shows the effectiveness of combining RSM, GA, and PSO for accurate prediction and optimization of operating parameters in wastewater electrolysis for green hydrogen production. The identified optimal conditions contribute to enhanced efficiency and increased hydrogen yield, thereby promoting the advancement of sustainable energy systems.

1. Introduction

As the world population raises, economies expand and the consequences of fossil fuel consumption intensify, the global energy landscape faces significant challenges. For many decades, fossil fuels such as coal, oil, and natural gas have been the predominant energy sources. Concerns have been raised about their long-term viability and sustainability due to their limited supply, environmental impact, and contribution to climate change [1]. Alternative energy sources have emerged as the primary focus for addressing these issues. Several promising alternatives, such as biodiesel, biogas, and hydrogen, have emerged to replace or supplement fossil fuels. These alternatives provide numerous benefits, such as reduced greenhouse gas emissions, increased energy security, and sustainable resource utilization [2]. As a secondary energy source, hydrogen can be produced through a variety of processes, including natural gas steam reforming and electrolysis [3]. Historically, the production of hydrogen has relied heavily on fossil fuels, specifically natural gas, through a process known as steam methane reforming (SMR). However, the carbon emissions associated with SMR mitigate the environmental benefits of hydrogen as a fuel source [4]. Therefore, there is an urgent need to develop environmentally friendly and economically viable alternatives for hydrogen production. Electrolysis of water has emerged as a promising alternative method for producing hydrogen from renewable energy sources with high purity [5,6]. However, certain disadvantages must be recognized. The relatively high unit cost of the electrical energy required for the electrolysis process is one such disadvantage. In addition, the anode and cathode materials used in electrolysis systems are expensive and have durability concerns [7]. To address these issues, one potential solution would be to obtain the necessary electrical energy from renewable sources, such as solar or

* Corresponding author.
E-mail address: aqueelahmad191@gmail.com (A. Ahmad).

https://doi.org/10.1016/j.ijhydene.2024.01.302
Received 1 August 2023; Received in revised form 2 January 2024; Accepted 25 January 2024
Available online 6 February 2024

wind power. This would not only reduce the process's environmental impact but also increase its cost-effectiveness [8]. Additionally, investigating chemical modifications to the system's electrodes could mitigate overvoltage values and improve overall efficiency [9]. Furthermore, from a thermodynamic standpoint, hydrogen has approximately three times the energy content of fossil fuels. Hydrogen has the potential to meet the requirements of a greener, cleaner, and more sustainable energy future when all of these factors are considered [10]. Despite the benefits, the widespread adoption of HHO production through water electrolysis and solar power faces obstacles. Optimization of electrolysis parameters, such as current density, temperature, and electrode materials, to improve efficiency and maximize hydrogen gas yield is a significant obstacle. In order to achieve cost-effective and economically viable production, the price of solar panels, energy storage systems, and electrolysis equipment must also be considered [11,12]. To optimize the efficiency and HHO yield from wastewater, it is necessary to control a number of process parameters. These parameters include voltage, temperature, electrolysis time, and amount of catalyst [9]. Nevertheless, the complexity of the electrolysis process and the interaction between these parameters make it difficult to manually determine the optimal conditions. To address these difficulties, advanced optimization techniques such as Response Surface Methodology (RSM), artificial neural network (ANN), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and ANYSYS, among others, have been implemented [13–17]. RSM is a statistical modeling technique that is vital in modeling and optimizing complex processes by establishing relationships between the response variable, such as hydrogen yield, and a set of input variables known as process parameters [18]. RSM is a valuable tool for optimizing a vast array of fields and industries. It allows for the development, enhancement, and optimization of complex processes by identifying the optimal production parameters. The primary benefit of RSM is its capacity to reduce the number of experimental runs required to achieve statistically significant results [19]. Through graphical representations such as 3D response surface plots and 2D contour plots, RSM facilitates the visualization of the shape and characteristics of the response surface, thereby facilitating the comprehension and interpretation of output responses. GA and PSO are effective optimization techniques that are widely employed in a variety of fields, including the optimization of the hydrogen production process [20]. GA is an evolutionary algorithm for computation inspired by natural selection. It follows the genetic inheritance, mutation, and selection processes in order to efficiently find optimal solutions in complex problem spaces [21]. GA begins with a population of potential solutions and evolves them iteratively through selection, crossover, and mutation operations. GA identifies the most promising options and progressively refines them to find the global optimum by evaluating the fitness of each solution based on objective functions [22]. In contrast, PSO is inspired by the collective behavior of social organisms, such as flocks of birds and schools of fish. Maintaining a population of particles that traverse the solution space in search of optimal solutions is its mode of operation. Each particle adjusts its position and velocity based on its own experience and the data acquired from the best-performing particles in its immediate vicinity [23]. PSO converges to the optimal solution through this iterative process by exploiting promising regions of the search space while maintaining diversity [24]. GA and PSO both offer multiple benefits for process optimization. They can handle complex and multidimensional search spaces efficiently, enabling the identification of optimal values for process parameters that maximize hydrogen production or minimize energy consumption. Furthermore, these optimization techniques reduce the number of required experimental runs, saving expenditures, resources, and time [25]. Several studies have investigated the production of HHO through water electrolysis. Masjuki et al. [26] conducted an experimental study on alkaline water electrolysis using SS-316 L electrodes and KOH as the electrolyte. They found that increasing the number of electrodes resulted in a proportional increase in current flow. The optimal conditions for HHO gas production were achieved using a single anode and two cathodes, generating 2150 cc of gas in 15 min. Ezzahra et al. [27] focused on HHO generation through alkaline water electrolysis using different cathode materials. They discovered that Zn95 % Cr5 % and Zn90 % Cr10 % alloys exhibited higher HHO gas generation compared to other alloys under various operating parameters. Kowthaman et al. [28] employed RSM to optimize HHO gas generation. They used a micro-patterned graphite electrode and alkaline sodium hydroxide as the electrolyte. By analyzing the combined effects of voltage, electrolyte molarity, and electrode distance, they identified optimal conditions that resulted in a maximum HHO generation rate of 2.3 ml/min. Kady et al. [29] conducted a study on the use of 3-mm-thick SS-316 steel plates and NaOH and KOH electrolytes in hydroxy gas production using an electrolyzer. Their study revealed a positive correlation between voltage and electrolysis efficiency, with a notable increase in efficiency from 2 V to 4 V, reaching a maximum of 51.12 %. However, the efficiency declined beyond 4 V. In addition, they determined that the most optimal model had a concentration of 5 % and a current of 14 A. Sekar et al. [30] investigated the use of 316 steel sheets, 2 mm plate spacing, and sodium hydroxide electrolytes for hydroxy gas production in engine applications, achieving a production rate of 2.5 L/min. Budiman et al. [31] explored the impact of salt electrolytes on hydroxy gas production using a dry-type electrolyzer and demonstrated that NaCl application led to rapid hydrogen gas production. Shahabadi et al. [32] developed a multi-cell electrolyzer using 316 stainless steel sheets for HHO gas generation without a membrane, highlighting the importance of optimal selection of electrolyzer plates and solar panel surface area when integrating an HHO generator with a solar system. The optimized design involved 18, 17, and 15 plates for one, two, and three 325 W solar panels, respectively.

In light of the existing literature, green hydrogen generation through water electrolysis is highly appealing due to its environmentally friendly and sustainable nature. Nevertheless, the commercialization of green hydrogen encounters substantial obstacles as producers persistently endeavor to decrease production costs while upholding stringent quality standards, posing a persistent challenge. Efficiently identifying the optimal operating parameters is crucial, given the numerous parameters that can impact hydrogen generation yield. Currently, conducting extensive experimental tests is the sole approach to determining the optimal combination of these parameters. However, these tests require substantial financial and labor investments. Therefore, there is a pressing need for prediction and optimization methods that can effectively identify the optimal operating parameters. Based on our current understanding, there have been limited research efforts focused on the generation of green hydrogen through wastewater electrolysis. Furthermore, no studies have specifically investigated the optimization of the solar-powered electrolysis technique for efficient green hydrogen production using a hybrid optimization approach. Hence, the objective of this study is to fill this research gap by thoroughly investigating the influence of crucial process parameters, such as catalyst (KOH) amount (g), electrode voltage (V), and electrolysis time (min), on hydrogen yield (L/min) using a novel RSM-GA-PSO-coupled hybrid optimization approach. By exploring innovative methods to generate green hydrogen, which can be derived from a readily available and renewable resource like wastewater, the study contributes to the circular economy and promotes the efficient utilization of resources. This research has the potential to make a significant impact on society and the environment by offering a cleaner and more sustainable energy alternative while addressing the challenges associated with traditional hydrogen production methods.

2. Material and methodology

2.1. Design and manufacturing of HHO generator

This study involved the design and fabrication of a dry-cell HHO generator. The generator consisted of 34 electrode plates made of

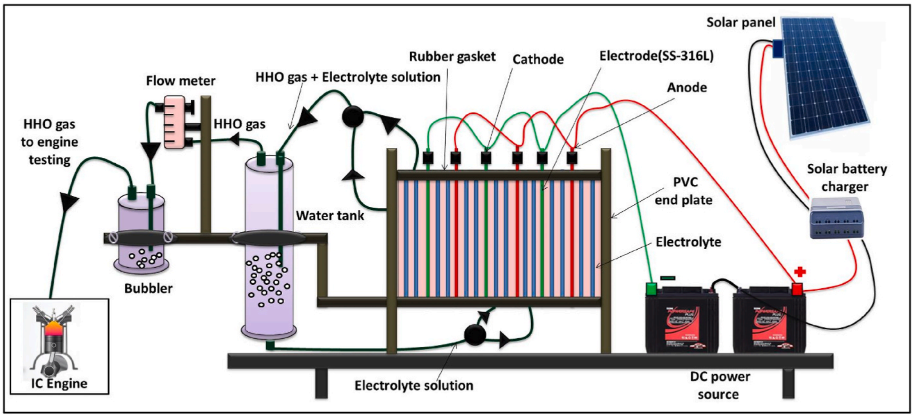

Fig. 1. Schematic of the dry cell HHO generator.

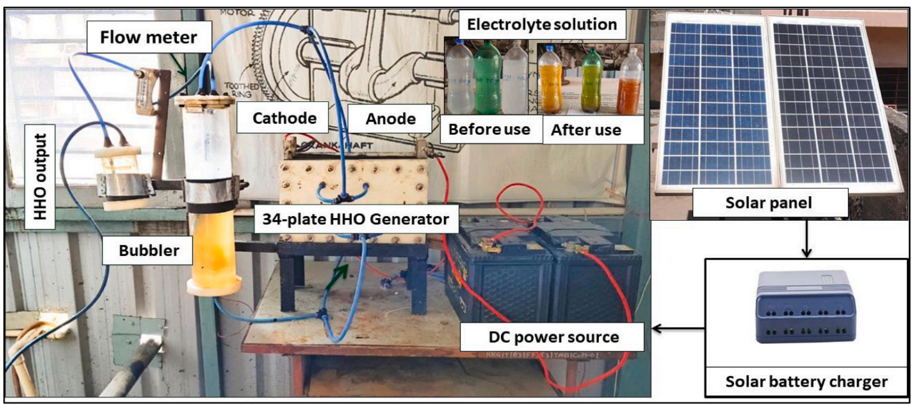

Fig. 2. Actual view of the experimental setup.

Table 1
Specification of electrolysis cell.

S No	Description	Material	Quantity	Specifications
1	Electrode plate	stainless steel-316 L	34	33 × 20 × 0.15 cm
2	Rubber gaskets	Ethylene propylene diene monomer	–	3.5 mm
3	End plate	PVC	2	40 × 25 × 4 cm
4	Electrolyte	KOH	(50-75-100 g)/2 L	–
5	Operating life	–	–	30,000 h

stainless steel (SS-316 L) with dimensions of 33 × 20 × 0.15 cm. Stainless steel 316 was selected as the electrode material due to its superior corrosion resistance in the presence of alkaline solutions [33]. To prevent contact between adjacent plates, corrosion-resistant rubber gaskets measuring 3.5 mm in thickness were utilized. These gaskets were composed of Ethylene Propylene Diene Monomer (EPDM). The assembly involved attaching three plates to the negative charge (cathode) and three plates to the positive charge (anode), while the remaining plates remained neutral. Notably, the top and bottom electrodes were equipped with four holes, each measuring 11 mm in diameter. The presence of holes at the bottom of the cell enables the passage of electrolytes between the plates, ensuring effective ion conduction within the cell. A continuous supply of electrolytes is necessary for optimal cell performance, as it not only facilitates ion conduction but also helps dissipate the heat generated during electrolysis [34]. These bottom holes in a dry cell allow the flow of electrolytes to each individual cell. As the oxyhydrogen gas produced within the cell has a lower density, it naturally rises. The holes located at the top of the cell serve the purpose of facilitating the passage of the oxy-hydrogen gas through the outlet pipe of the cell. To ensure the safety of the system, the HHO gas produced underwent a water trap process. The stainless steel 316 L grade electrolyzer demonstrates an operational lifespan of 30,000 h, signifying its robust durability and sustained functionality over an extended period of continuous operation [33]. Additional investigations demonstrate the identical life cycle of the electrolyzer made with SS-316 L [30,32]. Figs. 1 and 2 depict both a schematic and an actual view of the HHO generator. The alkaline electrolyte used in the generator was created by introducing KOH to wastewater, and the generator had a capacity of 2 L of electrolyte. The electrolysis process was conducted by employing DC current from a battery, which offered voltage options of 12, 18, or 24 V. The voltage and current intensity were measured using a Mastech MAS830L Multimeter. The battery itself was charged through a solar panel with a maximum power output of 250 W and a voltage of 30.8 V. Within the HHO generation unit, two water bubblers were utilized. One served the purpose of electrolyte recirculation, while the other was employed for safety reasons. The quantity of HHO gas produced was directed through the first bubbler, which was an acrylic pipe, and subsequently measured using a gas flow meter positioned before the second bubbler. Table 1 outlines its specifications.

2.2. *Water electrolysis mechanism*

Water electrolysis is a well-established and efficient process utilized for the generation of oxy-hydrogen (HHO) gas. This method is widely acknowledged for its simplicity and effectiveness. One of its notable advantages is the ability to produce HHO on demand, eliminating the

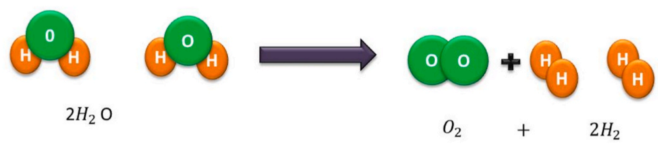

Fig. 3. HHO gas generation from water molecule decomposition.

Table 2
Physiochemical properties of hydrogen.

Properties	Unit	Parameter value of Produced HHO gas
Chemical formula	–	$H_2 + 0.5O_2$
Molecular weight	kg/kmol	2.016
Density @STD	Kg/m^3	0.083
Calorific Value	MJ/kg	119.87
Octane number	–	130
Auto ignition temperature	^{0}C	585
Minimum ignition energy	mJ	0.018

necessity for storage facilities and enhancing cost-effectiveness [4,7]. In this study, wastewater has been used for electrolysis. Wastewater has both positive and negative impacts on the electrolyzer's performance. On the positive side, it serves as a cost-effective source of water and contains ions that enhance electrolyte conductivity, leading to improved efficiency. Wastewater's organic compounds act as reducing agents, increasing hydrogen production rates. However, high ion concentrations cause corrosion, and organic compounds can lead to fouling, reducing catalytic activity [12]. Equations (1)–(8) represent the investigation of energy interactions and the thermodynamics associated with water electrolysis. The calculation of the change in free energy (ΔG) in a reaction involves subtracting the total reversible work (W_{rev}) achievable from the flow work ($P\Delta V$). This relationship is mathematically expressed by Equation (1):

$$-\Delta G = W_{rev} - P\Delta V \quad (1)$$

In the lack of flow work, Equation (1) can be simplified to:

$$-\Delta G = W_{rev} \quad (2)$$

The equilibrium cell voltage (E^0), also referred to as electromotive force, signifies the essential voltage threshold necessary for the reaction to take place. E^0 can be determined by calculating the disparity between the potentials of the anode and cathode.

$$E^0 = E^0_{anode} - E^0_{cathode} \quad (3)$$

In order to facilitate the movement of an electrical charge across a potential difference (E^0), a certain amount of energy must be expended. The amount of energy required for the transfer of charge can be determined by multiplying the number of charge transfers (n), the potential difference, and Faraday's constant (F). Hence, the equation representing the energy expended in transporting 'n' charges is nFE^0. Consequently, the electrical energy involved in the reaction and the total energy that can be obtained from the reaction, disregarding any changes in volume, can be expressed as:

$$W_{rev} = nFE^0 \quad (4)$$

By substituting W_{rev} into Equation (1), the expression for ΔG becomes:

$$-\Delta G = nFE^0 \quad (5)$$

The overall standard potential (E^0) is obtained by summing the potentials of the individual half-reactions at the anode and cathode.

The cathode reaction: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ (−0.83v) (6)

The Anode reaction: $2OH^- \rightarrow 0.5O_2 + H_2O + 2e^-$ (−0.4V) (7)

Overall reaction: $H_2O \rightarrow H_2 + 0.5O_2$ (−1.23V) (8)

These equations are applicable under standard conditions of 1 bar and 298 K (STP). Under these conditions, the electrolysis of water requires a theoretical potential difference of 1.23 V, resulting in the production of HHO gas [35].

2.3. Analysis of HHO gas

The efficiency of the HHO generator was assessed by determining the ratio between the energy generated by the HHO gas and the power consumed during its production. The HHO gas is composed of 2 mol of hydrogen and 1 mol of oxygen, as depicted in Fig. 3. Consequently, the mole fractions of hydrogen and oxygen in the gas are 0.66 and 0.33, respectively. This implies that hydrogen constitutes 66 % of the volume of the HHO gas ($V_{H2} = 0.66 \times V_{HHO}$). Since only H_2 gas is flammable, the efficiency of the HHO generator was computed using Equation (9) [36].

$$\eta_{HHO} = \frac{V_{H2} \times \rho_{H2} \times LHV_{H2}}{V \times I \times t} \quad (9)$$

Where V_{H2} denotes the volume of hydrogen (m^3) in the HHO gas. LHV_{H2} represents the lower heat value of hydrogen, which is equivalent to 119.87 MJ/kg. Additionally, ρ_{H2} signifies the density of hydrogen gas under standard conditions, which is 0.083 kg/m^3.The variables V, I, and t represent the measured experimental values for voltage (volts), current (amps), and time (seconds), respectively. A thorough examination of the physiochemical properties of HHO gas, a gas mixture composed of hydrogen (H_2) and oxygen (O_2) produced by water electrolysis, was carried out. The weighted average of the molecular weights of H_2 and O_2 was used to calculate the molecular weight of HHO gas. The density of HHO gas was calculated under standard conditions using the ideal law. Furthermore, the calorific value of HHO gas was measured and reported in MJ/kg, illuminating important information about its energy content. However, the properties such as octane number, auto ignition temperature (^{0}C), flame velocity (m/s), and minimum ignition energy (mJ) are traditionally applicable to liquid fuels, and their relevance to gaseous HHO was carefully evaluated. Throughout the investigation, careful consideration was taken to ensure the safety of procedures and methodology, and expert consultations were sought to validate the accuracy and reliability of the data obtained for this unique and less standardized gas mixture. Table 2 contains a summary of all the comprehensive findings.

2.4. Soft computing modeling technique

This study employed a Box-Behnken Design (BBD) based on RSM to accurately model and predict the HHO gas generation through wastewater electrolysis. An analysis of variance (ANOVA) was used to determine the strong connections between the input parameters (catalyst quantity, electrode voltage, and electrolysis time) of the electrolysis process and the resulting response (HHO yield). The desirability technique was used to improve the control parameters to achieve an optimal response. Furthermore, a novel genetic algorithm-particle swarm optimization (GA-PSO) technique was used to identify the ideal values of the input parameters for electrolysis, thereby attaining the most favorable response. Fig. 4 provides a detailed representation of the proposed modeling and optimization technique.

2.5. Response Surface Methodology (RSM)

RSM is a systematic and analytical approach employed to optimize process parameters to maximize or minimize the response variable. In the case of HHO generation, the response variable is the HHO yield (L/min). RSM is useful as it allows the experimenter to identify the optimal values of process parameters with fewer experimental runs. The process of RSM involves building a model that links the response variable to the

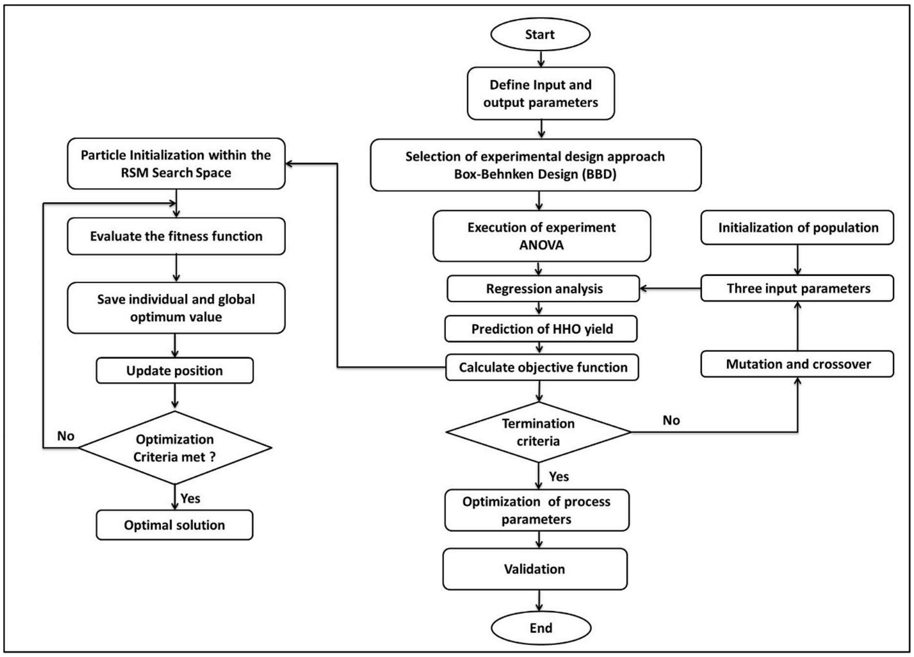

Fig. 4. Flow chart of the RSM-GA-PSO approach.

Table 3
Control variables and their levels.

Independent Variable	Coded level		
	−1	0	1
Catalyst amount (g):A	50	75	100
Electrode voltage (V):B	12	18	24
Electrolysis time (min):C	10	20	30

Table 4
L_{17} RSM-BBD experimental design matrix.

Run	Catalyst amount (g) (A)	Electrode voltage (V) (B)	Electrolysis time (min) (C)	Actual HHO (L/ min)	Model Predicted HHO (L/ min)
1	75	18	20	7.1	7.28
2	50	18	30	8	7.8
3	100	18	30	9.5	9.38
4	75	18	20	7.2	7.28
5	75	18	20	7.4	7.28
6	75	12	30	7.2	7.41
7	75	12	10	5.5	5.39
8	50	24	20	9.5	9.59
9	50	12	20	5.7	5.69
10	75	24	10	10.3	10.09
11	100	24	20	12.3	12.31
12	75	18	20	7.5	7.28
13	100	18	10	8	8.2
14	50	18	10	5.3	5.42
15	75	24	30	11.5	11.61
16	100	12	20	7.4	7.31
17	75	18	20	7.2	7.28

input factors using regression analysis. The model represented as a mathematical equation that describes the relationship between the variables. The model was employed to determine the response variable for given combination of input constraints. Following are the important RSM steps.

Step 1: Identify the process parameters that affect the response variable: In the case of HHO generation, process factors such as the catalyst amount (g), electrode voltage (V), and electrolysis time (min) can affect the HHO yield. The codes, ranges, and levels of several independent variables are presented in Table 3.
Step 2: Design a series of experiments: Experiments are designed to collect data on the response variable and the input variables. For this experiment, an L_{17} orthogonal array was developed using BBD at three factors and three levels, as shown in Table 4.
Step 3: Fit a mathematical model: The Design Expert® tool (version 8.0.6) was used to simulate the experiment data generated from the electrolysis process, which corresponds to the nonlinear multivariable regression equation Eq. (10). The model was used to determine the response variable for any given combination of input variables.

$$Y = \beta 0 + \sum_{i=1}^{3} \beta ixi + \sum_{i=1}^{3} \beta iixi^2 + \sum\nolimits_{i=1}^{2} \sum_{J=i+1}^{3} \beta ijxixj, + \mathcal{E} \tag{10}$$

Y is an output variable, i.e., HHO yield (L/min), xi and xj are input variables, β∘, βi, βii, and βij are the regression coefficients and ε is the model error.

Step 4: Analyze the results: The data obtained from the experiment was subjected to analysis using statistical software in order to identify the factors that exhibit a significant impact on the HHO yield. An analysis of variance (ANOVA) was employed to ascertain the primary effects and interactions of each factor.
Step 5: Optimize the process: On the basis of the experimental findings, the optimal conditions to attain maximum HHO yield have been identified. The RSM numerical optimization feature was used to model and determine the optimum parameters for generating HHO in wastewater electrolysis.

2.6. Data preprocessing

The model was developed using a dataset obtained from an HHO gas generator. Before conducting any analysis, the data underwent preprocessing, specifically descriptive statistical analysis, to gain insights into its characteristics [37]. Skewness is a statistical measure that quantifies the asymmetry of a property's probability density function

Table 5
Descriptive statistical analyses of the data.

Parameters	Mean	SD	Sum	Minimum	Median	Maximum	Skewness	Kurtosis
Catalyst amount (g)	75	17.6776	1275	50	75	100	0	−0.7428
Electrode voltage (V)	18	4.2426	306	12	18	24	0	−0.7428
Electrolysis time (min)	20	7.071	340	10	20	30	0	−0.7428
HHO yield (L/min)	8.0352	1.9849	136.6	5.3	7.4	12.3	0.7637	0.09807

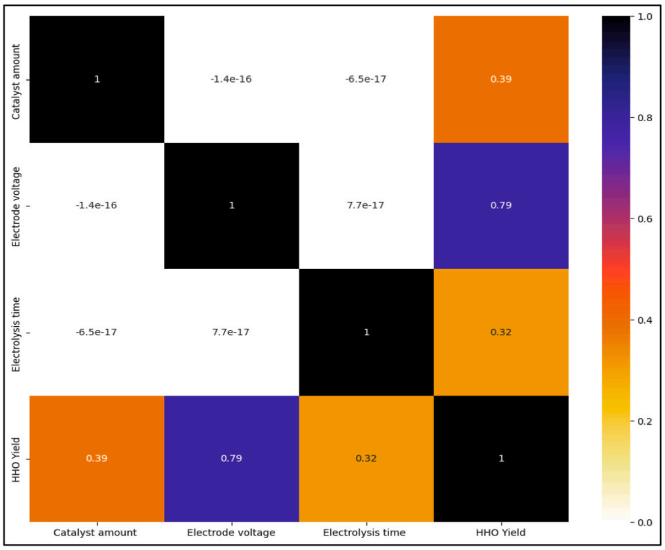

Fig. 5. Heat map of the correlation coefficient.

around its mean. A skewness value of zero indicates a property with a normal distribution, while positive or negative values indicate asymmetry in the probability density function. A positive skewness value suggests that the left side of the function is larger than the right side, indicating a greater number of low-value data points compared to high-value data points. Conversely, a negative skewness value implies a similar relationship in the opposite direction [38]. The analysis of the data shows that the majority of the input data had skewness values close to or equal to zero, indicating a normal distribution. Additionally, the output data exhibited a skewness value close to zero. Kurtosis, on the other hand, is a statistical measure that compares the shape of the data distribution to that of a normal distribution. In a normal distribution, the kurtosis value is zero. A flatter distribution yields a negative kurtosis value, while a more peaked distribution yields a positive kurtosis value. Furthermore, a descriptive statistical analysis of the data is presented in Table 5, and a heatmap in Fig. 5 illustrates the correlation coefficient between the input and output data, also known as the Pearson correlation coefficient. This coefficient matrix incorporates the electrolysis control parameters and the response variable. According to the heatmap, HHO yield exhibited the strongest linear relationship with electrode voltage (0.79), followed by catalyst amount (0.39) and electrolysis time (0.32).The correlation matrix depicted in Fig. 6 provides a visual representation of the intricate relationships among multiple factors that significantly impact the HHO yield. This study offers a comprehensive analysis of the data distribution, employing scatter plots and histograms to visually depict the variables under investigation. Furthermore, the investigation includes an assessment of the correlation coefficient, which offers valuable insights into the interrelationships between the variables and their associations with other factors. The dataset utilized in this study exhibited symmetrical characteristics and demonstrated the reliability and robustness of the findings.

2.7. Uncertainty assessment of HHO generation unit

Uncertainty analysis of an HHO gas generation unit, including the

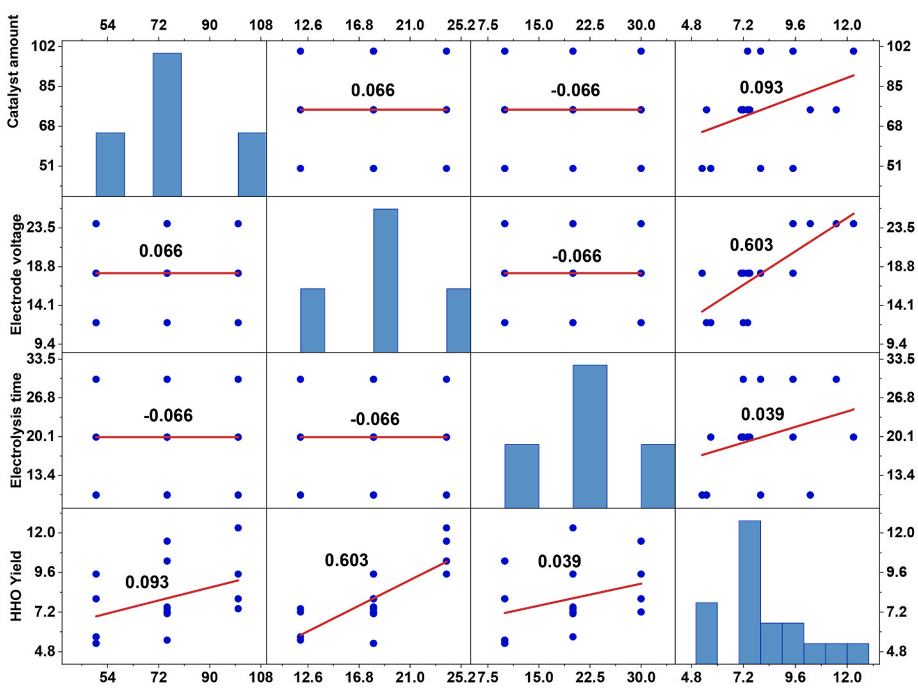

Fig. 6. Correlation matrix.

Table 6
ANOVA outcomes for Quadratic Model.

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F
Model	62.6883	9	6.9653	139.1086	**< 0.0001 significant**
A-Catalyst amount	9.4612	1	9.4612	188.955	<0.0001
B-Electrode voltage	39.605	1	39.605	790.97	<0.0001
C-Electrolysis time	6.3012	1	6.3012	125.8452	<0.0001
AB	0.3025	1	0.3025	6.0413	0.0436
AC	0.36	1	0.36	7.1897	0.0315
BC	0.0625	1	0.0625	1.2482	0.3008
A^2	0.2846	1	0.2846	5.6845	0.0486
B^2	5.9125	1	5.9125	118.081	<0.0001
C^2	0.1077	1	0.1077	2.1527	0.1858
Residual	0.3505	7	0.05007		
Lack of Fit	0.2425	3	0.0808	2.9938	**0.1585 not significant**
Pure Error	0.108	4	0.027		
Cor Total	63.0388	16			
Std. Dev.	0.22	R-Squared			0.9944
Mean	8.04	Adj R-Squared			0.9872
C.V. %	2.78	Pred R-Squared			0.9357
Adeq Precision	40.351	PRESS			4.05

power source and flow meter, entails quantifying the measurement and operation uncertainties associated with these components. Typically, the method entails identifying the pertinent parameters, determining their measurement uncertainties, conducting an error propagation analysis, and taking into account calibration, environmental conditions, and system variability [8]. The uncertainty analysis was performed for the main components of the HHO gas generator, including the voltmeter, Ammeter, flow meter, and solar power charging unit. This analysis is crucial because it provides a comprehensive understanding of the potential errors and variations in the gas generation process, allowing for an accurate evaluation of the system's performance, dependability, and efficiency [39]. It aids in identifying critical parameters, optimizing operations, establishing confidence in measurements, and making accurate choices regarding maintenance, calibration, or system enhancements. Uncertainty analysis ensures that reported results or measurements come with a confidence level, improving the overall dependability and credibility of the performance evaluation of the HHO gas generation unit. The combined standard uncertainty was calculated using the root-sum-of-squares method as depicted in eq. (11).

$$\begin{aligned} \textit{Total uncertainities} = \sqrt{}\{ & (\textit{uncertainity of } \text{ HHO ggHO eras flow meter})^2 + (\textit{uncertainity of } \text{voltmeter})^2 + (\textit{uncertainity of Stop watch})^2 \\ & + (\textit{uncertainity of } \text{solar pannel})^2 + (\textit{uncertainity of } \text{ammeter})^2 \\ & + (\textit{uncertainity of } \text{solar battery cctery hargerbolar tern unitented in Figure 2etup})^2 \\ & + (\textit{uncertainity of digital } \text{weight machinebolar tern unitented in Figure 2etup})\} \end{aligned} \tag{11}$$

$$= \sqrt{\{(0.48)^2 + (0.41)^2 + (0.2)^2 + (5)^2 + (0.41)^2 + (1.20)^2 + (0.5)^2\}}$$

$$= \pm 5.22\%$$

3. Results and discussion

3.1. Identification and analysis of model

The analysis of HHO generation (L/min) was conducted using the Design Expert software. The ratio between the highest yield (12.3 L/min) and the lowest yield (5.3 %) was determined to be 2.32. This ratio, which is less than 10, suggests that no modifications were necessary for the model to achieve the desired response. A quadratic function was selected as the model to determine the optimal conditions for HHO generation. The appropriateness of the quadratic model was evaluated by assessing the coefficient of determination (R^2) and performing the Fischer test to analyze the model's goodness of fit. The statistical significance of the model was assessed by examining the F-value and *P*-value [40].The purpose of the ANOVA analysis conducted on HHO yield, as shown in Table 6, was to determine and evaluate the most suitable model for accurately predicting the optimal conditions for HHO generation. The model exhibited a significant F-value of 139.1086, indicating its significance. The probability of noise being the cause of such an F-value at this level is 0.01 %. Model terms with *P*-values below 0.05 are considered significant, while those exceeding 0.1 are deemed statistically insignificant [41]. In this case, the terms A, B, C, AB, AC, A^2, and B^2 are significant. To ensure the reliability of the model, a lack-of-fit analysis was also performed. The lack-of-fit F-value of 2.99 and the p-value of 0.1585 were observed to be not statistically significant compared to the pure error. This indicates that the model fits the dataset well and provides accurate results.

The model's overall predictability was assessed using the R^2 metric. A value closer to 1 indicates a higher level of accuracy in predicting experimental results, whereas a value closer to zero indicates potential errors in the model's estimation of experimental data [42]. The study yielded an R^2 value of 0.9944, indicating a high level of predictability. Furthermore, the adjusted R^2 value was determined to be 0.9873, whereas the predicted R^2 value was estimated to be 0.9358. The slight discrepancy observed between these values shows a strong alignment between the model and experimental results, particularly in relation to the expected yield data. The coefficient of variation (CV) quantifies the relative percentage change in the standard deviation with respect to the mean value. A CV value of 5 % is considered advantageous. In the present study, the model demonstrated a high level of predictability, as evidenced by a low CV value of 2.78 %. This finding suggests that the predictions were highly accurate [39]. In addition, the Adeq Precision value, which assesses the model's sensitivity by quantifying the signal-to-noise ratio, was calculated to be 40.351. The observed value exceeds the established threshold of 4, which signifies a favorable level of accuracy and validates the model's effectiveness in predicting hydrogen production efficiency. The predicted residual sum of squares (PRESS) is a statistical metric used to evaluate the predictive performance of a model. This metric calculates the total squared differences between the predicted and actual values of the response variable for each observation in the dataset. The study demonstrates that a PRESS value of 4.05 signifies a superior predictive fit of the model to the data [14]. Equation (12) represents the recommended software equations for predicting HHO yield. Fig. 7(a) depicts a comparison between the actual and predicted values of HHO, showing minimal disparity and accurate predictions by the quadratic function. Fig. 7(b) presents a residual vs. run plot, which indicates that the residuals are randomly distributed, suggesting constant variance across all tests. To ascertain the optimal transformation for HHO yield, a Box-Cox plot (Fig. 7(c)) was constructed, depicting the natural logarithm of the residual plotted against time. The green line in the plot represents the lambda (ƛ) value, indicating the minimum residual. Notably, the software suggests a lambda value of 1 as the recommended transformation.

$$\text{HHO Yield}\left(\frac{\text{L}}{\text{min}}\right) = 9.137 - 0.0279\text{A} - 0.91\text{B} + 0.15225\text{C} + 0.00183\text{AB} - 0.0012\text{AC} - 0.00208\text{BC} + 0.000416\text{A}^2 + 0.0329\text{B}^2 + 0.0016\text{C}^2 \quad (12)$$

3.2. Effect of process parameters on HHO gas generation

The effect of process parameters on HHO gas generation was investigated using three independent variables: catalyst amount, electrode voltage, and electrolysis time. These variables were analyzed to determine their influence on the dependent variable, HHO gas yield. The study followed the three steps of RSM: experimental design, data collection, and the construction of predictive models based on the parameters. The predicted results obtained from the model equation for HHO generation were compared with the actual test results, and a good agreement was observed, as shown in Fig. 7(a). This demonstrated the reliability of the model in predicting the HHO generation rate (L/min). To better understand the effects of the independent variables on the desired response, 3D and 2D response surface graphs were generated based on Eq. (12), which represents the amount of HHO yield. The 3D and 2D response surface graphs for catalyst amount, electrode voltage, and electrolysis time are presented in Fig. 8. ANOVA Table 6 indicated that the electrode voltage (F-value 790.97) had a more significant effect on the amount of hydrogen production compared to the catalyst amount (F-value 188.955) and electrolysis time (F-value 125.8452). In addition to the individual effects of the parameters, significant interaction effects were observed among the process parameters. The interaction between catalyst amount vs. electrode voltage and catalyst amount vs. electrolysis time had a synergistic effect on HHO gas generation, further enhancing the HHO gas yield. Fig. 8(a and b) depicts the surface and contour plots between catalyst amount and electrode voltage when electrolysis time held a constant value of 20 min. It was observed that increasing the catalyst amount from 50 g to 100 g led to an increase in hydrogen production. Additionally, increasing the electrode voltage from 12 V to 24 V resulted in higher hydrogen production when the electrolysis time was fixed. For instance, when the electrolysis time was set at 20 min, an electrode voltage of 12 V yielded a HHO gas amount of 7.4 L/min, whereas an electrode voltage of 24 V resulted in a HHO gas amount of 12.31 L/min with a catalyst amount of 100 g. This shows the dominant effect of electrode voltage on the process. Fig. 8(c and d) demonstrated that varying the catalyst amount from 50 g to 100 g and the electrolysis time from 10 min to 30 min significantly impacted the amount of hydrogen production when the electrode voltage was fixed at 18 V. Similarly, Fig. 8(e and f) illustrated that increasing the applied potential value in the electrolysis system led to an increase in hydrogen production. The hydrogen volume continuously increased across all potential values from 12 V to 24 V. When the electrolysis time was set to 30 min, the hydrogen yield obtained with a catalyst amount of 75 g was 7.2 L/min and 11.5 L/min for applied potential values of 12 V and 24 V, respectively. These findings show the effects of the independent variables on HHO gas generation. The electrode voltage was found to have a substantial influence, followed by the catalyst amount and electrolysis time. The response surface graphs provided visual representations of these effects and further supported the conclusions drawn from the data. Overall, the results demonstrated that catalyst concentration, electrode voltage, and electrolysis time significantly influenced HHO gas generation. Increasing the catalyst concentration, electrode voltage, and electrolysis time led to higher HHO gas yields. The interaction effects among these parameters further enhanced the gas yield.

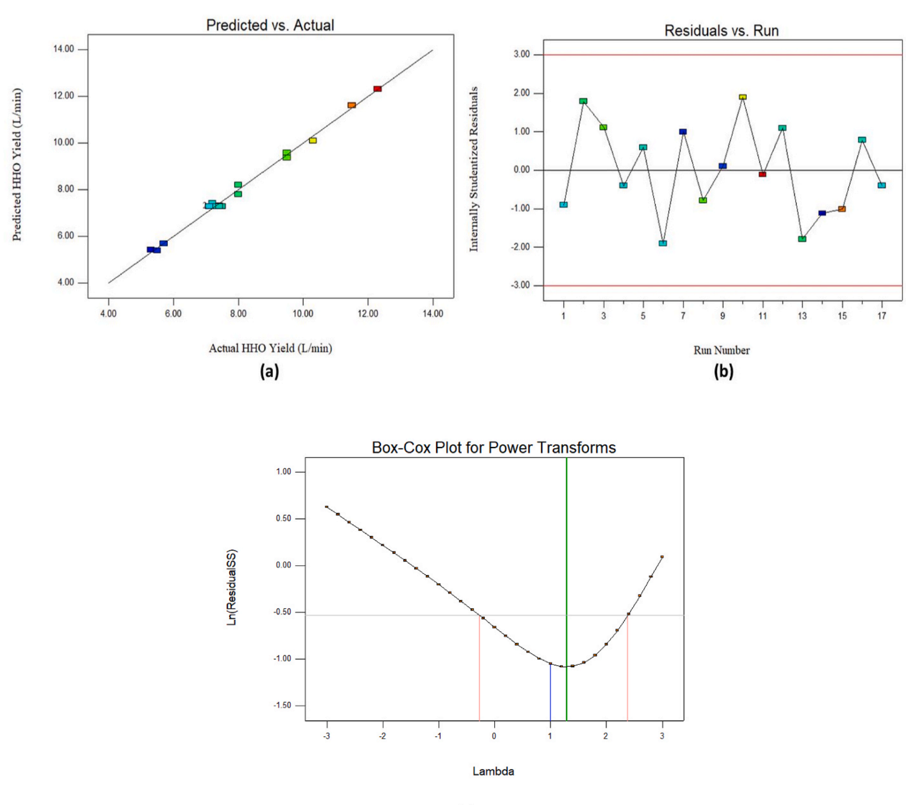

Fig. 7. (a) Predicted verses actual HHO yield (b) Residual verses Run response for HHO yield (c) Box-cox plot for power transforms.

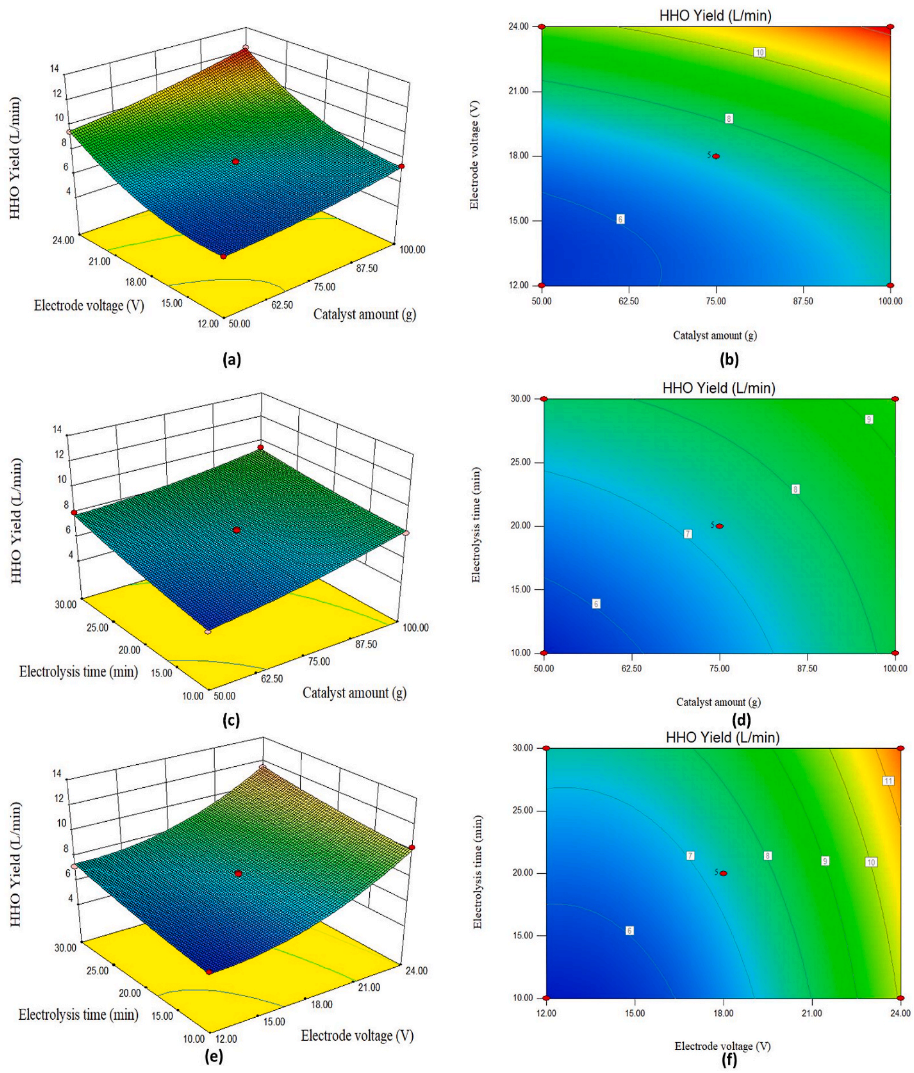

Fig. 8. Surface and contour plot for the effect of (a and b) catalyst amount and electrode voltage (c and d) catalyst amount and electrolysis time (e and f) electrode voltage and electrolysis time on HHO yield.

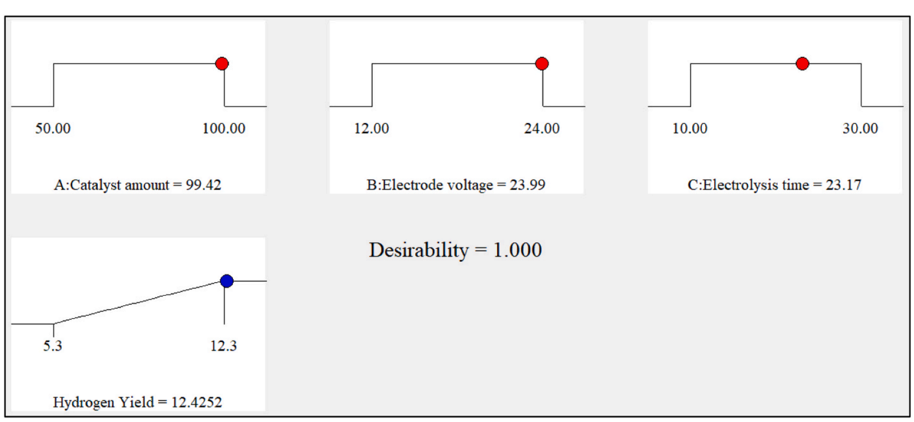

Fig. 9. Numerical Optimization parameters for HHO gas generation.

3.3. Optimization of process parameters

3.3.1. RSM desirability approach

This study investigated different optimization approaches offered by the DOE software to maximize the HHO gas generation. Among these methods, numerical optimization, renowned for its success in continuous optimization, was selected to identify the optimal combination of input parameters that would yield the maximum HHO gas production. The optimization process involved varying the amounts of catalyst, electrode voltages, and electrolysis times to achieve the optimal

Table 7
GA parameter settings.

$$\text{Fitness Function} = -(9.137 - 0.0279X1 - 0.91X2 + 0.15225X3 + 0.00183X1X2 - 0.0012X1X3 - 0.00208X2X3 + 0.000416X1^2 + 0.0329X2^2 + 0.0016X3^2) \quad (13)$$

Parameter	Values
Population type	Double Vector
Population size	50
Scaling Function	Rank
Selecting function	Stochastic Uniform
Creating function	Constraint dependent
Elite count	0.05*population size
Crossover fraction	0.8
Direction	Forward
Stopping criteria: generation	(100*Number of variables) = 300
Stopping criteria: Fitness limit	Infinite
Stopping criteria: Time	infinite

response. By utilizing the desirability function, an optimal value of the HHO gas yield was attained. The desirability function, encompassing multiple responses and optimization goals, provided a comprehensive approach to determining the highest achievable HHO yield [43]. According to the desirability function, the maximum HHO gas yield was found to be 12.42 L/min. This optimal yield was obtained at specific conditions: a catalyst amount of 99.42 g, an electrode voltage of 23.99 V, and an electrolysis time of 23.17 min, yielding a desirability value of 1.000, as shown in Fig. 9. Confirmatory experiments were performed using the optimal parameter combinations obtained through the RSM desirability approach. These validation experiments demonstrated the accuracy of the predictive model and the effectiveness of the optimization process [44]. Through the optimization approaches employed in this study, the ideal conditions for Hydrogen generation were determined, leading to a maximum HHO gas yield of 12.2 L/min. The error rate between the predicted and observed values was found to be only 1.62 %, demonstrating a close correlation between the expected and actual results. The close agreement between predicted and observed values further validates the accuracy and reliability of the developed predictive model. The combination of numerical optimization and the desirability function proved to be a robust and effective method for identifying the optimal process parameters.

3.3.2. Genetic algorithm (GA) approach

The GA is a heuristic optimization algorithm inspired by the process of natural selection. Known for its capability to tackle complex problems efficiently, GA has emerged as a powerful tool for finding optimal solutions [45]. In the context of optimizing HHO gas yield through wastewater electrolysis, a challenging and multifaceted problem arises involving the optimization of various factors, including catalyst amounts, electrode voltages, and electrolysis times. This study proposes the utilization of a GA to address the optimization of HHO yield. The primary objective was to identify the optimal conditions for the electrolysis reaction, maximizing the HHO gas yield. The GA operates by generating a population of potential solutions, where each solution represents a unique set of decision variables. These decision variables encompass catalyst amounts, electrode voltages, and electrolysis times. The implementation of the GA was carried out using the MATLAB software toolbox. Table 7 shows GA factor settings used in this analysis. The objective function for the GA was set as the HHO yield, and the algorithm proceeded through several generations, with each generation building upon the improvements made in the previous iteration. Through this iterative process, the GA aimed to converge towards the optimal solution that maximizes the HHO gas yield. To guide the GA in its search for the optimal solution, a mathematical equation was employed as the input parameter, and a quadratic RSM was employed to model the response surface. The RSM was based on BBD response values, calculated at a 95 % confidence level. Design Expert 8.0.7.1 software facilitated the generation of several equations, one of which is represented as Equation (12). This equation predicted experimental zone response values, aiding in the optimization process. The GA iteratively determines minimum and maximum values for the input process parameters, with specific GA parameters such as mutation rate, generations, population size, and cross-over rate playing essential roles in shaping the optimization process. The fitness functions, described by

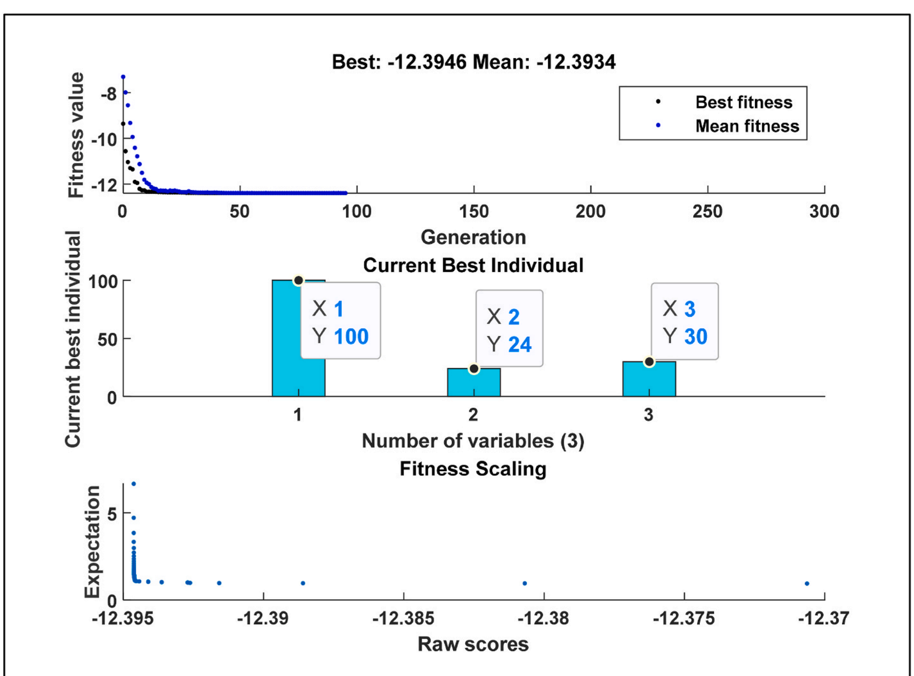

Fig. 10. Results of best fitness, mean fitness and Current best individual.

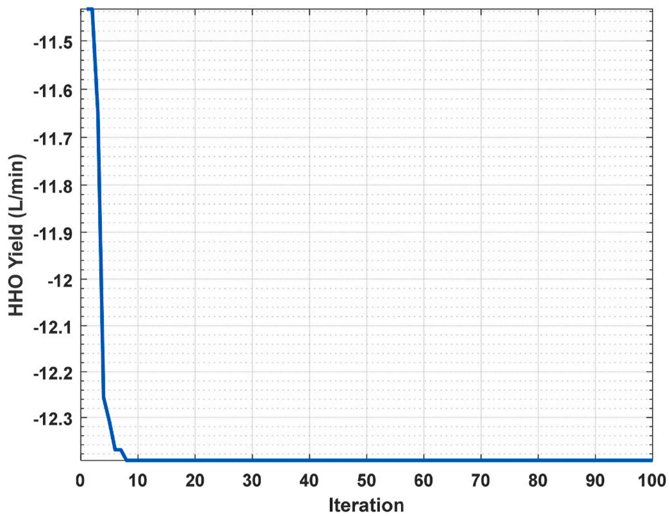

Fig. 11. Convergence of PSO for HHO yield.

Equation (13), guide the GA in evaluating and comparing potential solutions. Throughout the optimization process, the GA generates best and mean value, depicted in Fig. 10, which assist in assessing the performance of the algorithm. Moreover, the GA identifies the best individuals, representing optimal sets of decision variables that yield the maximum HHO gas. The findings demonstrate that the GA successfully determined the optimal conditions for wastewater electrolysis, resulting in the maximization of the HHO yield. The optimized settings identified by the GA were as follows: a catalyst amount of 100 g, an electrode voltage of 24 V, and an electrolysis time of 30 min. Under these optimized conditions, the attained HHO yield was measured at 12.39 L/min. Where fitness function is HHO yield, X_1 is catalyst amount, X_2 is electrode voltage, and X_3 is electrolysis time.

3.3.3. *Particle swarm optimization (PSO) approach*

In this study, the PSO algorithm, a widely recognized metaheuristic optimization technique, was used to maximize the HHO generation yield. The optimal process parameters for maximizing the HHO yield through electrolysis include catalyst amount, electrode voltage, and electrolysis time. The PSO algorithm was implemented using the MATLAB 2020a code, and after 20 iterations, the algorithm converged, determining the optimal parameter values. The maximum HHO gas yield achieved through PSO was 12.37 L/min, with a catalyst amount of 100 g, electrode voltage of 24 V, and electrolysis time of 30 min as the optimum input variables. To ensure accurate prediction of the HHO yield, the experiments were conducted three times, each time using different optimum input process variables. The fitness value versus the number of iterations is depicted in Fig. 11.

3.4. *Experimental validation*

To validate the optimization results and assess the accuracy of the predictions, experimentation was undertaken, and independent variables were identified for the RSM, GA, and PSO techniques. By utilizing the optimal parameter combinations attained through each optimization method, conclusive findings were obtained, verifying the effectiveness of the predictions. Through the implementation of multiple optimization approaches, the ideal production conditions for HHO gas generation were determined. Table 8 presents the optimal input process factors predicted by RSM, GA, and PSO for achieving a maximum HHO yield of 12.42 L/min, 12.39 L/min, and 12.37 L/min, respectively. Based on the specific goals of the study, the benefits of employing RSM for HHO gas generation through wastewater electrolysis became evident. RSM demonstrated superior performance by achieving a HHO yield of 12.42 L/min in 23.17 min, which is 22.76 % less time compared to the GA and PSO approaches. This reduction in computation time highlights the efficiency of the RSM method in identifying the optimal conditions for HHO gas production.

4. Conclusion

This work shows the successful implementation of hybrid statistical techniques to determine the optimal operating parameters for maximizing hydrogen production through wastewater electrolysis. The optimized parameters obtained through RSM outperformed the GA and PSO approaches, resulting in a higher HHO gas production rate of 12.42 L/min. The optimized operating parameters were determined as a catalyst amount of 99.42 g, an electrode voltage of 22.9 V, and an electrolysis time of 23.17 min. Notably, the use of RSM led to a time savings of 6.83 min compared to the GA and PSO approaches, indicating its efficiency in providing quicker optimization results. The ANOVA analysis shows significant interactions between catalyst amount and electrode voltage, as well as catalyst amount and electrolysis time, while other interactions had no significant effect on the HHO yield. Furthermore, solar energy was used for the electrolysis process, enhancing the sustainability and environmental benefits of the wastewater electrolysis system. This integration highlights the potential for renewable energy in HHO generation from wastewater.

Future research should focus on refining and optimizing the electrolysis process to enhance efficiency and hydrogen production. This can be achieved by exploring different electrolytes, electrode materials, and system configurations to improve overall performance. The identified optimal conditions provide a solid foundation for scaling up the wastewater electrolysis system for practical applications. Integration of the optimized process into larger-scale systems or wastewater treatment plants should be explored to assess its viability and potential for commercial implementation. Additionally, conducting economic evaluations and life cycle assessments will provide valuable insights into the economic feasibility and environmental impact of the optimized wastewater electrolysis process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 8
validation test.

Method	Catalyst Amount(g)	Electrode Voltage(v)	Electrolysis time (min)	Predicted HHO Yield (L/min)	Experimental HHO Yield (L/min)	Error (%)
RSM	99.42	23.99	23.17	12.42	12.1	2.57
GA	100	24	30	12.39	11.8	4.7
PSO	100	24	30	12.37	11.6	6.2

References

[1] Hoang Anh Tuan, Ong Hwai Chyuan, Fattah IM Rizwanul, Chong Cheng Tung, Cheng Chin Kui, Sakthivel R, Yong Sik Ok. Progress on the lignocellulosic biomass pyrolysis for biofuel production toward environmental sustainability. Fuel Process Technol 2021;223:106997.

[2] Abas Naeem, Kalair A, Khan Nasrullah. Review of fossil fuels and future energy technologies. Futures 2015;69:31–49.

[3] Hoang Anh Tuan, Huang ZuoHua, Nižetić Sandro, Pandey Ashok, Nguyen Xuan Phuong, Luque Rafael, Ong Hwai Chyuan, Said Zafar, Tri Hieu Le. Characteristics of hydrogen production from steam gasification of plant-originated lignocellulosic biomass and its prospects in Vietnam. Int J Hydrogen Energy 2022;47(7):4394–425.

[4] Kanimozhi B, Alsehli Mishal, Elfasakhany Ashraf, Veeman Dhinakaran, Balaji Santhosh, Kumar TR Praveen, Sekar Manigandan. Effects of oxyhydrogen on the CI engine fueled with the biodiesel blends: a performance, combustion and emission characteristics study. Int J Hydrogen Energy 2022;47(88):37668–76.

[5] Subramanian Balaji, Ismail Saleel. Production and use of HHO gas in IC engines. Int J Hydrogen Energy 2018;43(14):7140–54.

[6] Subramanian Balaji, Thangavel Venugopal. Analysis of onsite HHO gas generation system. Int J Hydrogen Energy 2020;45(28):14218–31.

[7] Nabil Tamer, Khairat Dawood Mohamed M. Enabling efficient use of oxy-hydrogen gas (HHO) in selected engineering applications; transportation and sustainable power generation. J Clean Prod 2019;237:117798.

[8] Bilbao Diego Contreras. Valorization of the waste heat given off in a system alkaline electrolyzer-photovoltaic array to improve hydrogen production performance: case study Antofagasta, Chile. Int J Hydrogen Energy 2021;46(61):31108–21.

[9] El Kady MA, Ahmed El Fatih Farrag, Gad MS, El Soly AK, Abu Hashish HM. Parametric study and experimental investigation of hydroxy (HHO) production using dry cell. Fuel 2020;282:118825.

[10] Gu Xiaosong, Cheng Mingliang, Zhang Xiaofeng, Zeng Yian. The pollutant discharge improvement by introducing HHO gas into biomass boiler. Int J Hydrogen Energy 2021;46(45):23292–300.

[11] El Soly AK, El Kady MA, Farrag Ahmed El Fatih, Gad MS. Comparative experimental investigation of oxyhydrogen (HHO) production rate using dry and wet cells. Int J Hydrogen Energy 2021;46(24):12639–53.

[12] Salek Farhad, Zamen Mohammad, Hosseini Seyed Vahid, Babaie Meisam. Novel hybrid system of pulsed HHO generator/TEG waste heat recovery for CO reduction of a gasoline engine. Int J Hydrogen Energy 2020;45(43):23576–86.

[13] Yahya, Mohidin Hamdya Sabrina, Abbas Tariq, Saidina Amin Nor Aishah. Optimization of hydrogen production via toluene steam reforming over Ni–Co supported modified-activated carbon using ANN coupled GA and RSM. Int J Hydrogen Energy 2021;46(48):24632–51.

[14] Secer Acelya, Hasanoğlu Arif. Evaluation of the effects of process parameters on co–gasification of Çan lignite and sorghum biomass with response surface methodology: an optimization study for high yield hydrogen production. Fuel 2020;259:116230.

[15] Munusamy Thurga Devi, Chin Sim Yee, Rahman Khan Md Maksudur. Optimization of process parameters for photoreforming of hydrogen evolution via response surface methodology (RSM): a study using Carbon@ exfoliated g–C3N4. Chem Eng Res Des 2022;177:513–25.

[16] Ahmad A, Yadav AK, Singh A. Process optimization of spirulina microalgae biodiesel synthesis using RSM coupled GA technique: a performance study of a biogas-powered dual-fuel engine. Int J Environ Sci Technol 2023:1–20.

[17] Ahmad Aqueel, Yadav Ashok Kumar, Singh Achhaibar, Singh Dinesh Kumar, Ağbulut Ümit. A hybrid RSM-GA-PSO approach on optimization of process intensification of linseed biodiesel synthesis using an ultrasonic reactor: enhancing biodiesel properties and engine characteristics with ternary fuel blends. Energy 2023:129077.

[18] Sharma Prabhakar. Prediction-optimization of the effects of di-tert butyl peroxide-biodiesel blends on engine performance and emissions using multi-objective response surface methodology. J Energy Resour Technol 2022;144(7):072301.

[19] Ahmad Aqueel, Ashok Kumar Yadav, Singh Achhaibar, Singh Dinesh Kumar. A comprehensive machine learning-coupled response surface methodology approach for predictive modeling and optimization of biogas potential in anaerobic Co-digestion of organic waste. Biomass Bioenergy 2024;180:106995.

[20] Naik Ravindra, Sathisha N. Desirability function and GA-PSO based optimization of electrochemical discharge micro-machining performances during micro-channeling on silicon-wafer using mixed electrolyte. Silicon 2022;14(15):10007–21.

[21] Khan Tahir Ali, Khan Tasmeem Ahmad, Yadav Ashok Kumar. A hydrodynamic cavitation-assisted system for optimization of biodiesel production from green microalgae oil using a genetic algorithm and response surface methodology approach. Environ Sci Pollut Control Ser 2022;29(32):49465–77.

[22] Alam Shahriar Tanvir, Tomal ANM Amanullah, Khan Nayeem Moddassir. High-speed machining of Ti–6Al–4V: RSM-GA based optimization of surface roughness and MRR. Results in Engineering 2023;17:100873.

[23] Sharafi Masoud, Tarek Y, Mekkawy EL. Multi-objective optimal design of hybrid renewable energy systems using PSO-simulation based approach. Renew Energy 2014;68:67–79.

[24] Bemani Amin, Xiong Qingang, Baghban Alireza, Habibzadeh Sajjad, Amir H, Mohammadi, Doranehgard Mohammad Hossein. Modeling of cetane number of biodiesel from fatty acid methyl ester (FAME) information using GA-, PSO-, and HGAPSO-LSSVM models. Renew Energy 2020;150:924–34.

[25] Ghorbani Narges, Kasaeian Alibakhsh, Toopshekan Ashkan, Bahrami Leyli, Amin Maghami. Optimizing a hybrid wind-PV-battery system using GA-PSO and MOPSO for reducing cost and increasing reliability. Energy 2018;154:581–91.

[26] Masjuki HH, Ruhul AM, Mustafi Nirendra N, Kalam MA, Arbab MI, Fattah IM Rizwanul. Study of production optimization and effect of hydroxyl gas on a CI engine performance and emission fueled with biodiesel blends. Int J Hydrogen Energy 2016;41(33):14519–28.

[27] Chakik Fatima Ezzahra, Kaddami Mohammed, Mikou Mohammed. Effect of operating parameters on hydrogen production by electrolysis of water. Int J Hydrogen Energy 2017;42(40):25550–7.

[28] Kowthaman CN, Senthil Kumar P, Arul Mozhi Selvan V. Micro-patterned graphite electrodes: an analysis and optimization of process parameters on hydrogen evolution in water electrolysis. Fuel 2021;305:121542.

[29] El Kady MA, Farrag Ahmed El Fatih, Gad MS, El Soly AK, Abu Hashish HM. Parametric study and experimental investigation of hydroxy (HHO) production using dry cell. Fuel 2020;282:118825.

[30] Sekar Dhileepan, Venkadesan Gnanamoorthi, Samuel Panithasan Mebin. Optimisation of dry cell electrolyser and hydroxy gas production to utilise in a diesel engine operated with blends of orange peel oil in dual-fuel mode. Int J Hydrogen Energy 2022;47(6):4136–54.

[31] Budiman Arif, Yerizam Muhammad, Bow Yohandri. Design of dry cell HHO generator using NaCl solution for hydrogen production. IJFAC (Indonesian Journal of Fundamental and Applied Chemistry) 2022;7(1):8–15.

[32] Shahdabadi Reza Shahraki, Zamen Mohammad, Hosseini Seyed Vahid. Adaptive design for the connection of multicell HHO generator with solar photovoltaic panels. Int J Hydrogen Energy 2023;48(86):33422–33.

[33] Said Zafar, Le Duc Trong Nguyen, Sharma Prabhakar, Dang Viet Ha, Le Huu Son, Nguyen Dinh Tuyen, Em Bui Thi Anh. Optimization of combustion, performance, and emission characteristics of a dual-fuel diesel engine powered with microalgae-based biodiesel/diesel blends and oxyhydrogen. Fuel 2022;326:124987.

[34] Trofimova, Tina-Tini S, Ostanina Tatjana N, Nikitin Vjacheslav S, Valentin M, Rudoi, Ostanin Nickolaj I, Trofimov Aleksej A. Modeling of the porous nickel deposits formation and assessing the effect of their thickness on the catalytic properties toward the hydrogen evolution reaction. Int J Hydrogen Energy 2021;46(32):16857–67.

[35] Ferrero D, Lanzini A, Santarelli M, Leone P. A comparative assessment on hydrogen production from low-and high-temperature electrolysis. Int J Hydrogen Energy 2013;38(9):3523–36.

[36] Kassaby M El, Eldrainy YA, Khidr ME, Khidr KI. Effect of hydrogen (HHO) gas addition on gasoline engine performance and Emmission. Alex Eng J 2016;55:243–51.

[37] Said Z, Sharma P, Tiwari AK, Huang Z, Bui VG, Hoang AT. Application of novel framework based on ensemble boosted regression trees and Gaussian process regression in modelling thermal performance of small-scale Organic Rankine Cycle (ORC) using hybrid nanofluid. J Clean Prod 2022;360:132194.

[38] Liao W, Luo Y, Chen T. Thermal-hydraulic performance analysis of outward convex corrugated tubes based on skewness and kurtosis. Int J Therm Sci 2021;165:106970.

[39] Das Samar, Kashyap Debangsu, Bora Bhaskor J, Kalita Pankaj, Kulkarni Vinayak. Thermo-economic optimization of a biogas-diesel dual fuel engine as remote power generating unit using response surface methodology. Therm Sci Eng Prog 2021;24:100935.

[40] Sharma Prabhakar, Sahoo Bibhuti B, Said Zafar, Hadiyanto H, Xuan Phuong Nguyen, Nižetić Sandro, Huang Zuohua, Anh Tuan Hoang, Li Changhe. Application of machine learning and Box-Behnken design in optimizing engine characteristics operated with a dual-fuel mode of algal biodiesel and waste-derived biogas. Int J Hydrogen Energy 2023;48(18):6738–60.

[41] Das Sutapa, Goud Vaibhav V. RSM-optimised slow pyrolysis of rice husk for bio-oil production and its upgradation. Energy 2021;225:120161.

[42] Tripathi Manoj, Bhatnagar Ashish, Nabisab Mujawar Mubarak, Sahu Jaya Narayan, Ganesan P. RSM optimization of microwave pyrolysis parameters to produce OPS char with high yield and large BET surface area. Fuel 2020;277:118184.

[43] Özgür Ceyla, Mehmet Erman Mert. Prediction and optimization of the process of generating green hydrogen by electrocatalysis: a study using response surface methodology. Fuel 2022;330:125610.

[44] Laougé, Boubacar Zakari, Çığgın Aslı Seyhan, Merdun Hasan. Optimization and characterization of bio-oil from fast pyrolysis of Pearl Millet and Sida cordifolia L. by using response surface methodology. Fuel 2020;274:117842.

[45] Fayyazi Ebrahim, Ghobadian Barat, Najafi Gholamhassan, Hosseinzadeh Bahram. Genetic algorithm approach to optimize biodiesel production by ultrasonic system. Chem Prod Process Model 2014;9(1):59–70.